**6** $p^3 \times p^x = p^9$

(a) Find the value of $x$.

Multiplying powers with the same base: add exponents.

3 + x = 9, so x = 9 − 3 = 6.

$x =$ 6

**1 mark**
Correct exponent

**(1)**

$(7^2)^y = 7^{10}$

(b) Find the value of $y$.

Raising a power to a power: multiply exponents.

2y = 10, so y = 10 ÷ 2 = 5.

$y =$ 5

**1 mark**
Correct exponent

**(1)**

$100^a \times 1000^b$ can be written in the form $10^w$

(c) Show that $w = 2a + 3b$

$100 = 10^2$ and $1000 = 10^3$

**1 mark**
Write each base as a power of 10

$100^a \times 1000^b = (10^2)^a \times (10^3)^b$

$= 10^{2a} \times 10^{3b} = 10^{2a + 3b}$ so $w = 2a + 3b$.

**1 mark**
Multiply exponents, then add them to complete the proof

**(2)**

**(Total for Question 6 is 4 marks)**

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